SC7 Bayes Methods MT26

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Lecture 9 : Approximate Bayesian Computation (ABC) and simulation-based inference*.

*Methods avoiding likelihood evaluation, just observation model simulations.

ABC - why work hard to fit a model that is wrong anyway?

We use ABC when the likelihood is intractable, but the generative model

$$(\theta, y) \sim \pi(\theta)p(y|\theta)$$

is easy to simulate. Arises in two cases.

(1) If $p(y|\theta) \propto \tilde{p}(y, \theta)$ (as a function of y) then

$$p(y|\theta) = \tilde{p}(y, \theta)/c(\theta)$$

with an intractable normalising constant depending on θ,

$$c(\theta) = \int \tilde{p}(y, \theta)dy.$$

(2) However, $p(y|\theta)$ may only be *defined by a big simulator*: we dont even have $\tilde{p}(y, \theta)$. This is the main motivation.

These problems are *doubly intractable*. MCMC doesnt work*.

*...well, not simple MCMC - the exchange variant of MCMC handles case (1)

ABC example: the Ising model - a doubly intractable model.

Let $\mathcal{Y} = \{0,1\}^{m^2}$ be all binary images $Y = (Y_1, Y_2, ..., Y_{m^2})$, $Y_i \in \{0,1\}$, with cell index $i = 1, ..., m^2$ on a square lattice.

$\#y$ counts disagreeing neighbors in image $Y = y$.

The *Ising model* is the following distribution over $\mathcal{Y}$:

$$p(y|\theta) = \exp(-\theta \#y)/c(\theta).$$

Here $\theta \geq 0$ is a positive *smoothing parameter* and

$$c(\theta) = \sum_{y \in \mathcal{Y}} \exp(-\theta \#y)$$

is a normalizing constant we cant easily* compute for m large.

*A formula for $c(\theta)$ was given in 2017. It's a great piece of work, but costly to evaluate and a small modification to the law for Y breaks the method.

Observe image data $y_{obs} \in \mathcal{Y}$ and seek estimate of θ.

If $\pi(\theta)$ is a prior for θ then the posterior is

$$\pi(\theta|y_{obs}) = \frac{p(y_{obs}|\theta)\pi(\theta)}{p(y_{obs})}.$$

Consider MCMC targeting $\pi(\theta|y_{obs})$ with a simple θ-proposal, say $\theta' \sim U(\theta - a, \theta + a),\ a > 0$. Reject if $\theta' < 0$ and otherwise,

$$\begin{aligned}
\alpha(\theta'|\theta) &= \min\left\{1,\ \frac{p(y_{obs}|\theta')\pi(\theta')}{p(y_{obs}|\theta)\pi(\theta)}\right\} \\
&= \min\left\{1, \frac{c(\theta)}{c(\theta')} \times e^{(\theta-\theta')\#y_{obs}} \times \frac{\pi(\theta')}{\pi(\theta)})\right\}
\end{aligned}$$

and although $p(y_{obs})$ cancels, $c(\theta)/c(\theta')$ does not, and so we are left with an acceptance probability we cannot evaluate.

Rejection sampling, ABC-style

The basic ABC algorithm approximates the following variant of rejection. Suppose y and θ are both discrete so $p(y|\theta) \leq 1$.

The following simulates $\Theta \sim \pi(\theta|y_{obs})$ where (as usual)

$$\pi(\theta|y_{obs}) \propto p(y_{obs}|\theta)\pi(\theta).$$

(0) Set $n = 0$.

(1) $n \leftarrow n + 1$. Simulate $\theta_n \sim \pi(\cdot)$ and $y_n \sim p(\cdot|\theta_n)$.

(2) If $y_n = y_{obs}$ return $\Theta = \theta_n,\ N = n$ and stop, else goto (1).

Given θ_n, line (2) succeeds w.p. $\Pr(Y = y_{obs}|\theta_n) = p(y_{obs}|\theta_n)$.

$\Rightarrow$ accept on a (1)-(2) pass w.p. $\sum_\theta \pi(\theta)p(y_{obs}|\theta) = p(y_{obs})$.

Claim: The return value $\Theta \sim \pi(\cdot|y_{obs})$.

Proof: To find the probability $\Theta = \theta$, sum over probabilities for all sequences of rejections ending in the return value θ,

$$\begin{aligned}
\Pr(\Theta_{\text{rej}} = \theta) &= \Pr(\Theta = \theta, N = 1) + \Pr(\Theta = \theta, N = 2) + ... \\
&= \pi(\theta)p(y_{obs}|\theta)(1 + r + r^2 + ...) \\
&= \pi(\theta)p(y_{obs}|\theta) \times \frac{1}{1 - (1 - p(y_{obs}))} \\
&= \pi(\theta|y_{obs})
\end{aligned}$$

since we reject at each pass over (1)-(2) w.p. $r = 1 - p(y_{obs})$.

Rejection-ABC (where now $(\Theta, Y) \in \Omega \times \mathcal{Y}$ - general rv).

If y is "close" to y_{obs} then $\pi(\theta|y)$ approximates $\pi(\theta|y_{obs})$.

Measure "close" using summary statistics $S(y) = (S_1(y), ..., S_p(y))$ chosen to inform θ.

Let $D(s, s_{obs})$ be distance between $s = S(y), s_{obs} = S(y_{obs})$.
Let $\delta > 0$ be a threshold distance.

(0) Set $n = 0$.

(1) $n \leftarrow n + 1$. Simulate $\theta_n \sim \pi(\cdot)$ and $y_n \sim p(\cdot|\theta_n)$.

(2) If $D(S(y), s_{obs}) < \delta$ stop with $\Theta = \theta_n,\ N = n$, else goto (1).

Claim: Define $\pi_{ABC}(\theta|y_{obs}) = \pi(\theta|Y \in \Delta_\delta(y_{obs})),\ \theta \in \Omega$.

The distribution of the return value is $\Theta \sim \pi_{ABC}(\cdot|y_{obs})$.

Proof: Take $A \in \mathcal{B}_\Omega$ and show $\Pr(\Theta \in A) = \pi_{ABC}(A|y_{obs})$.

Let $\Delta_\delta(y_{obs}) = \{y \in \mathcal{Y} : D(S(y), s_{obs}) < \delta\}$.

Accept at (1)-(2) with probability

$$p(\Delta_\delta(y_{obs})) = \int_{\Delta_\delta(y_{obs})} \int_\Omega \pi(\theta)\, p(y|\theta)\, d\theta dy.$$

$$\begin{aligned}
\Pr(\Theta \in A) &= \sum_{n=1}^{\infty} \Pr(\Theta \in A, N = n) \\
&= \sum_{n=1}^{\infty} \Pr(y_j \notin \Delta_\delta(y_{obs})\ \forall j < n,\ (\theta_n, y_n) \in A \times \Delta_\delta(y_{obs})) \\
&= \sum_{n=1}^{\infty} \Pr((\theta_n, y_n) \in A \times \Delta_\delta(y_{obs}))(1 - p(\Delta_\delta(y_{obs})))^{n-1} \\
&= \frac{\int_A \int_{\Delta_\delta(y_{obs})} p(y|\theta) dy \pi(\theta) d\theta}{p(\Delta_\delta(y_{obs}))} = \int_A \frac{p(\Delta_\delta(y_{obs})|\theta)\pi(\theta)d\theta}{p(\Delta_\delta(y_{obs}))} \\
&= \pi_{ABC}(A|y_{obs})
\end{aligned}$$

We have made two (more) approximations

- we summarise the data with $y_{obs} \rightarrow s(y_{obs})$

- we replace the data-statement

"the realised value of the data is $Y = y_{obs}$"

with

"the realised value of the data is in the ball $Y \in \Delta_\delta(y_{obs})$".

If $S : \mathcal{Y} \rightarrow R^p$ is a sufficient statistic then

$$\pi(\theta|Y = y_{obs}) = \pi(\theta|S(Y) = s(y_{obs})),$$

and in that case

$$\pi(\theta|Y \in \Delta_\delta(y_{obs})) \rightarrow \pi(\theta|y_{obs})$$

as $\delta \rightarrow 0$ may often be verified.

Example

Data model: $y_i \sim \text{Poisson}(\Lambda)^*$, $i = 1, 2, ..., n$ with $n = 5$.

Prior: $\Lambda \sim \Gamma(\alpha = 1, \beta = 1)$.

Summary statistic: $S(y) = \bar{y}$

Distance measure: $D(\bar{y}, \bar{y}_{obs}) = |\bar{y} - \bar{y}_{obs}|$ and $\delta = 0.5, 1$.

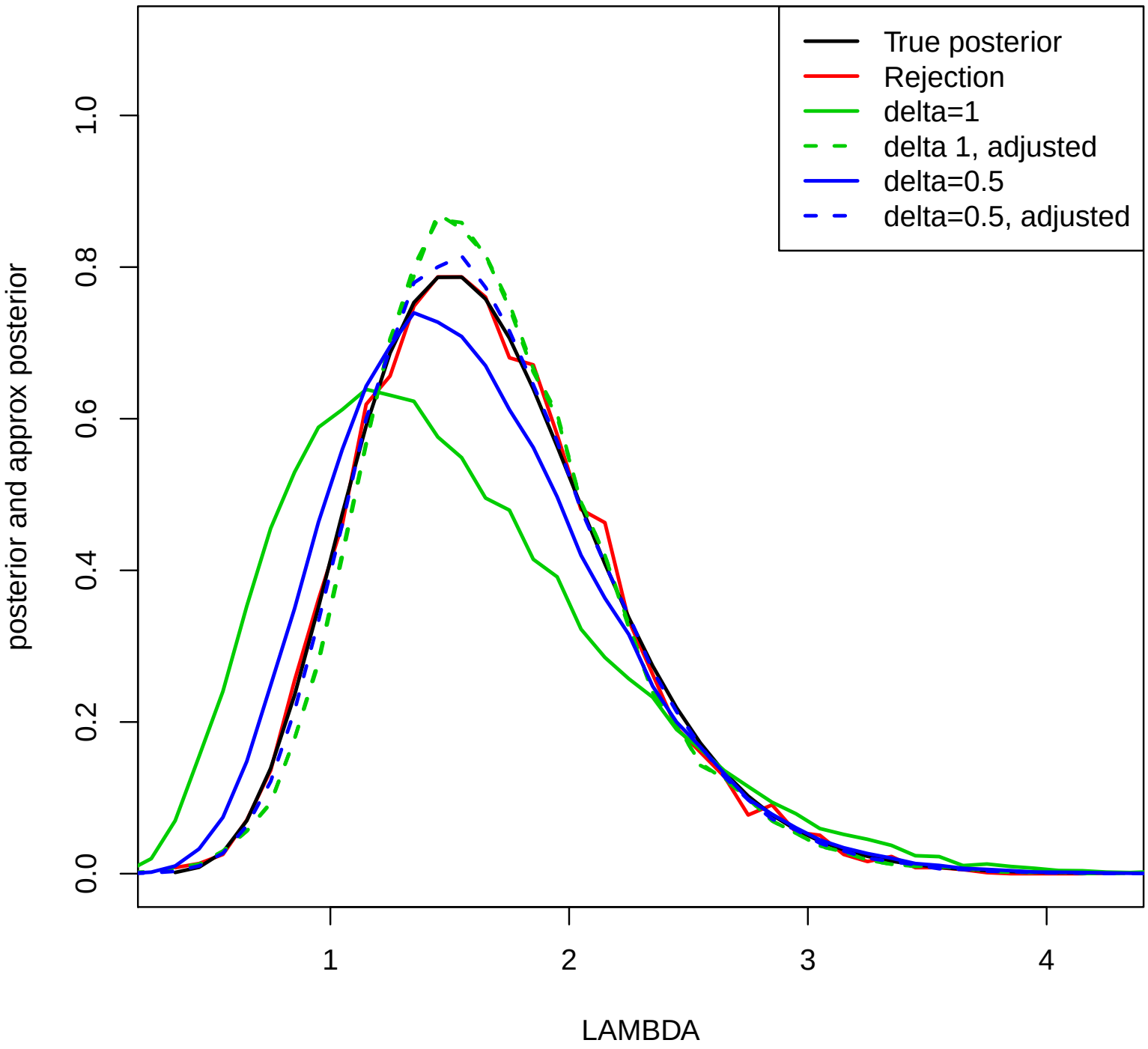
ABC algorithm

(1) Simulate $\lambda \sim \Gamma(\alpha, \beta)$ and $y_i \sim \text{Poisson}(\lambda), i = 1, 2, ..., n$.

(2) If $|\bar{y} - \bar{y}_{obs}| < \delta$ return λ and stop, otherwise goto (1).

We do Bayesian inference without calculating $p(y_{obs}|\theta)$ or $\pi(\theta)$, sometimes called "likelihood-free" or "simulation-based" inference. We just specify how to simulate parameters and data.

*True value was $\Lambda = 2$.


True posterior
Rejection
delta=1
delta 1, adjusted
delta=0.5
delta=0.5, adjusted
posterior and approx posterior
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
4
LAMBDA

Regression adjustment of samples

ABC generates $\theta, y \sim \pi(\theta)p(y|\theta)$. Conditional on y, $\theta \sim \pi(\theta|y)$. "Shift" this distribution to match $\pi(\theta|y_{obs})^*$.

Let $\theta \sim \pi(\cdot|y)$, $\theta' \sim \pi(\cdot|y_{obs})$, $s = S(y)$ and $s_{obs} = S(y_{obs})$.

Assume (1) The posterior mean $\mu(S(y)) = E(\theta|Y = y)$ is a linear function of $S(y)$ alone, that is, for some $\beta \in R^p$,

$$\mu(s) = \mu(s_{obs}) + (s - s_{obs})^T \beta.$$

Assume (2) The random variables $\theta - \mu(s)$ and $\theta' - \mu(s_{obs})$ are identically distributed.

These assumptions must hold for $y \in \Delta_\delta(y_{obs})$.

A1 is a local approximation good when $\mu(s)$ is smooth near $s_{\rm obs}$; A2 assumes little change in posterior shape over $\Delta_\delta(y_{obs})$.

*We set this out for a scalar θ. Multivariate θ is a straightforward extension.

Claim: Let $\theta_{adj} = \theta - (s - s_{obs})^T \beta$. If A1 & A2 then

$$\theta_{adj} \sim \pi(\cdot | y_{obs})$$

Proof: By A2 $\theta \sim \theta' + \mu(s) - \mu(s_{obs})$, so

$$\theta - (s - s_{obs})^T \beta \sim \theta' + \mu(s) - \mu(s_{obs}) - (s - s_{obs})^T \beta$$

and so by A1 $\theta - (s - s_{obs})^T \beta \sim \theta'$.

Claim: Let $\theta_{adj} = \theta - (s - s_{obs})^T\beta$. If A1 & A2 then

$$\theta_{adj} \sim \pi(\cdot|y_{obs})$$

Proof: By A2 $\theta \sim \theta' + \mu(s) - \mu(s_{obs})$, so

$$\theta - (s - s_{obs})^T\beta \sim \theta' + \mu(s) - \mu(s_{obs}) - (s - s_{obs})^T\beta$$

and so by A1 $\theta - (s - s_{obs})^T\beta \sim \theta'$.

Regression adjustment Let $\epsilon \sim \theta - \mu(s)$ so $\theta \sim \mu(s) + \epsilon$ and

$$\theta \sim \mu(s_{obs}) + (s - s_{obs})^T\beta + \epsilon \qquad \text{by A1.}$$

Estimate β by regressing θ on $(s - s_{obs})$:
(1) run ABC at fixed δ and get pairs of samples $(\theta^{(t)}, y^{(t)})_{t=1}^T$.
(2) Let $s^{(t)} = S(y^{(t)})$. Take

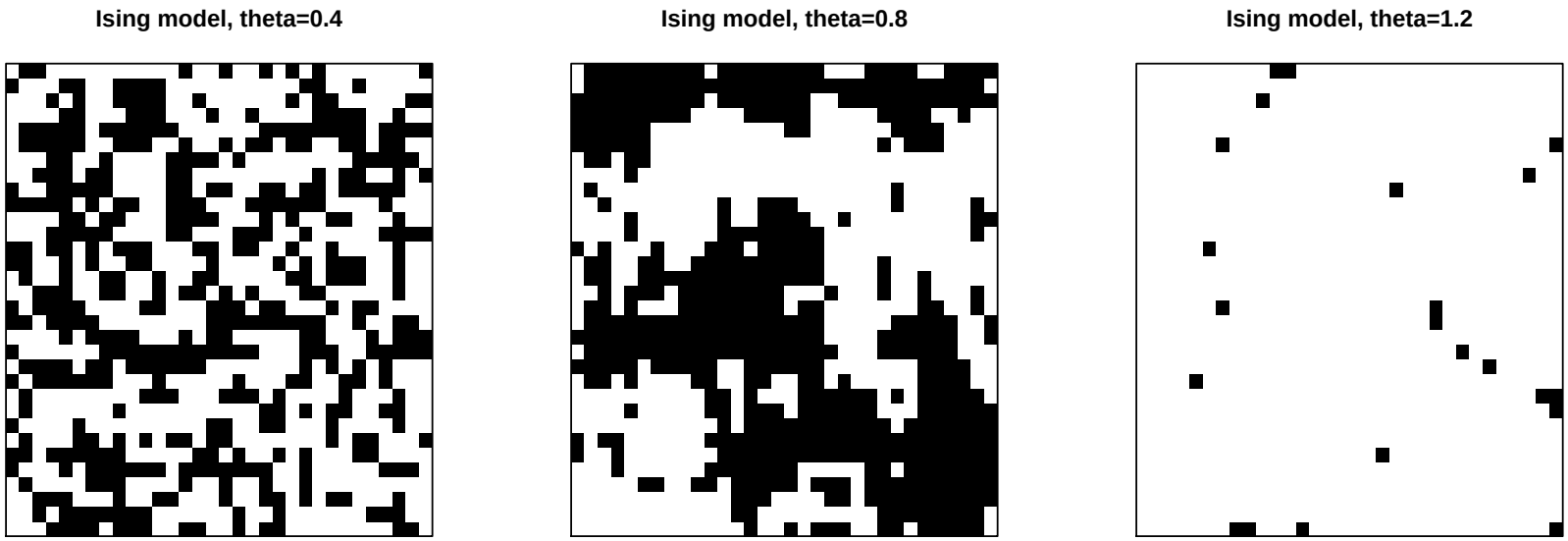
$$(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_{t=1}^T (\alpha + (s^{(t)} - s_{obs})^T\beta - \theta^{(t)})^2$$

If $\hat{\theta}_{adj}^{(t)} = \theta^{(t)} - (s^{(t)} - s_{obs})^T\hat{\beta}$ then

$$\hat{\theta}_{adj}^{(t)} \sim \pi(\cdot|y_{obs}) \quad \text{(approx).}$$

ABC example: the Ising Model

The Ising distribution $Y \sim p(\cdot|\theta)$ is easy to sample for moderate m-values using MCMC. Here are 3 samples $Y \sim p(\cdot|\theta)$:



These samples are not exactly distributed according to $p(\cdot|\theta)$ (convergence) but we can make them as good as we need by taking long MCMC runs.

ABC inference for θ for Ising model

Recall the doubly intractable inference for $\theta|y_{obs}$.

We have data $Y = y$ which is an $m \times m$ binary matrix and our observation model for Y, $Y \sim p(y|\theta)$ is the Ising model. Our goal is to estimate θ. The posterior is

$$\pi(\theta|y_{obs}) = \frac{p(y_{obs}|\theta)\pi(\theta)}{p(y_{obs})}$$

and the likelihood

$$p(y_{obs}|\theta) = \exp(-\theta \# y_{obs})/c(\theta)$$

depends on $c(\theta)$, a function of θ which is intractable here.

ABC algorithm sampling $\theta \sim \pi(\theta|y_{obs})$ approximately:

Prior $\theta \sim \mathrm{Exp}(2)$ has right scale.

Statistic: $S(y_{obs}) = \#y_{obs}$, (which is sufficient).

(1) Simulate $\theta \sim \mathrm{Exp}(2)$ and $y \sim \exp(-\theta \#y)/c(\theta)$ (MCMC).

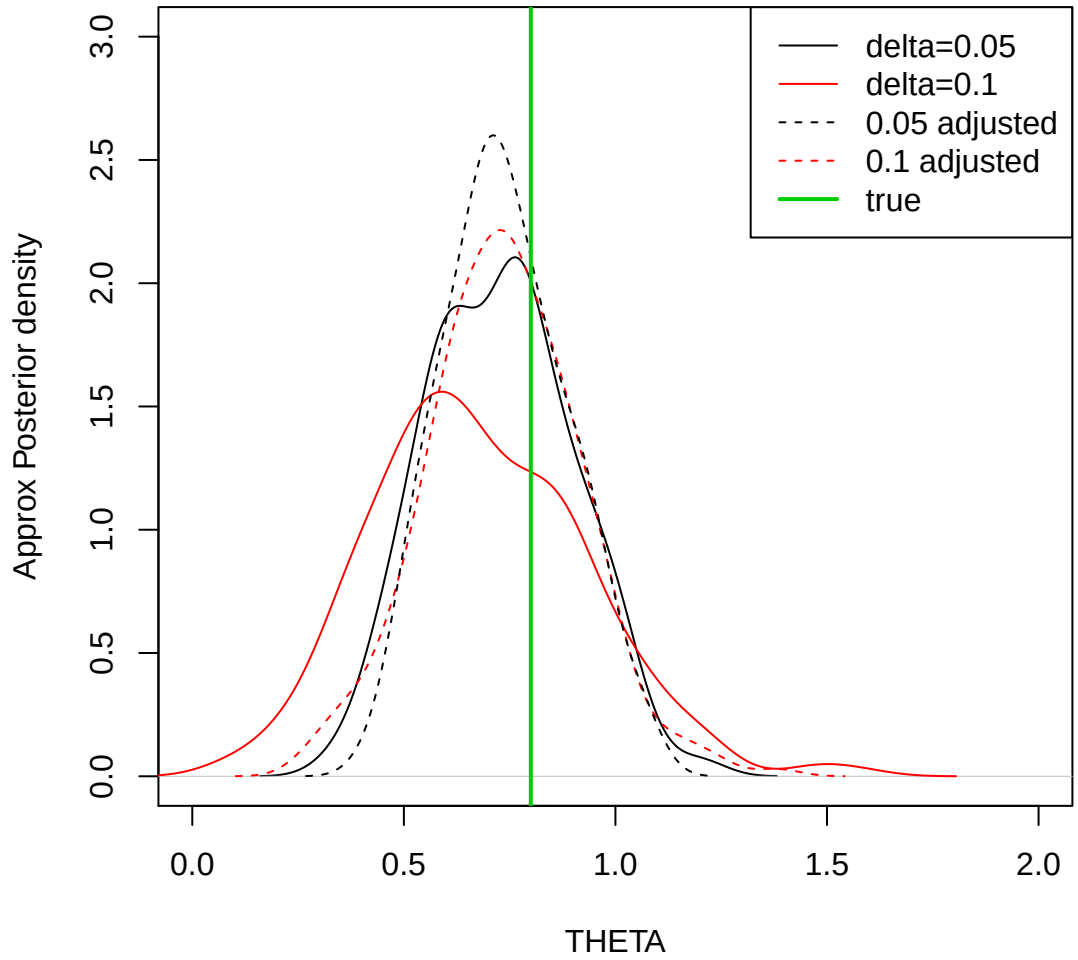
(2) If $|\#y - \#y_{obs}| < \delta \times 2m(m-1)^*$ return θ, else goto (1).

Implemented in online code: 8x8 Ising, true $\theta = 0.8$.

The ABC-posterior converges to true posterior as $\delta \to 0$.

The regression adjustment for $\delta = 0.1$ approximately corrects its distribution to better agree with that for $\delta = 0.05$.

*scale by $\max(\#y)$


delta=0.05
delta=0.1
0.05 adjusted
0.1 adjusted
true
Approx Posterior density
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0.0
0.5
1.0
1.5
2.0
THETA